Student Performance

Task

Perform a comprehensive analysis of student performance using the dataset "student_performance.csv". This involves data cleaning, exploratory data analysis (EDA) with univariate, bivariate, and multivariate visualizations, and building two predictive models: a Multinomial Logistic Regression model to classify performance into 'Low', 'Medium', and 'High' categories, and a Linear Regression model to predict final scores. Evaluate both models using appropriate metrics such as Accuracy, Confusion Matrix, R-squared, MAE, and MSE, and summarize the key factors influencing student success.

Project Setup and Data Loading

Subtask:

Define objectives, load the student performance dataset, and perform initial data inspection.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('student_performance.csv')

# Display the first few rows
print('First 5 rows of the dataset:')
display(df.head())

# Check data types and non-null counts
print('\nDataFrame Info:')
df.info()

# Summary statistics for numerical columns
print('\nSummary Statistics:')
display(df.describe())
```

```
First 5 rows of the dataset:
```

	StudentID	Name	Gender	AttendanceRate	StudyHoursPerWeek	PreviousGrade	ExtracurricularActivities	ParentalSupport
0	1.0	John	Male	85.0	15.0	78.0	1.0	High
1	2.0	Sarah	Female	90.0	20.0	85.0	2.0	Medium
2	3.0	Alex	Male	78.0	10.0	65.0	0.0	Low
3	4.0	Michael	Male	92.0	25.0	90.0	3.0	High
4	5.0	Emma	Female	NaN	18.0	82.0	2.0	Medium

```
DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 12 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   StudentID                  960 non-null    float64
 1   Name                       966 non-null    object
 2   Gender                     952 non-null    object
 3   AttendanceRate             960 non-null    float64
 4   StudyHoursPerWeek          950 non-null    float64
 5   PreviousGrade              967 non-null    float64
 6   ExtracurricularActivities  957 non-null    float64
 7   ParentalSupport            978 non-null    object
 8   FinalGrade                 960 non-null    float64
 9   Study Hours                976 non-null    float64
 10  Attendance (%)             959 non-null    float64
 11  Online Classes Taken       975 non-null    object
dtypes: float64(8), object(4)
memory usage: 93.9+ KB

Summary Statistics:
```

	StudentID	AttendanceRate	StudyHoursPerWeek	PreviousGrade	ExtracurricularActivities	FinalGrade	Study Hours	Atte
count	960.000000	960.000000	950.000000	967.000000	957.000000	960.000000	976.000000	959.
mean	5416.019792	85.510417	17.630526	77.598759	1.520376	80.030208	2.406967	77.
std	2653.748319	7.332125	6.272132	10.006640	1.046439	9.493652	1.620267	19.
min	1.000000	70.000000	8.000000	60.000000	0.000000	62.000000	-5.000000	50.
25%	3113.500000	82.000000	12.000000	70.000000	1.000000	72.000000	1.200000	63.
50%	5396.500000	88.000000	18.000000	78.000000	1.000000	80.000000	2.500000	76.
75%	7754.750000	91.000000	22.000000	86.000000	2.000000	88.000000	3.700000	89.

Data Cleaning and Wrangling

Subtask:

Clean the dataset by handling missing values, duplicates, and anomalous data, and prepare features for analysis.

Clean the dataset by removing duplicates, handling missing values with median/mode imputation, correcting anomalies in Study Hours and Attendance, and dropping non-predictive columns as instructed.

```
import pandas as pd

# 1. Remove duplicate rows
df.drop_duplicates(inplace=True)

# 2. Address missing values
# Numerical columns: impute with median
num_cols = df.select_dtypes(include=['float64', 'int64']).columns
for col in num_cols:
    df[col] = df[col].fillna(df[col].median())

# Categorical columns: impute with mode
cat_cols = df.select_dtypes(include=['object']).columns
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

# 3. Rectify data anomalies
```

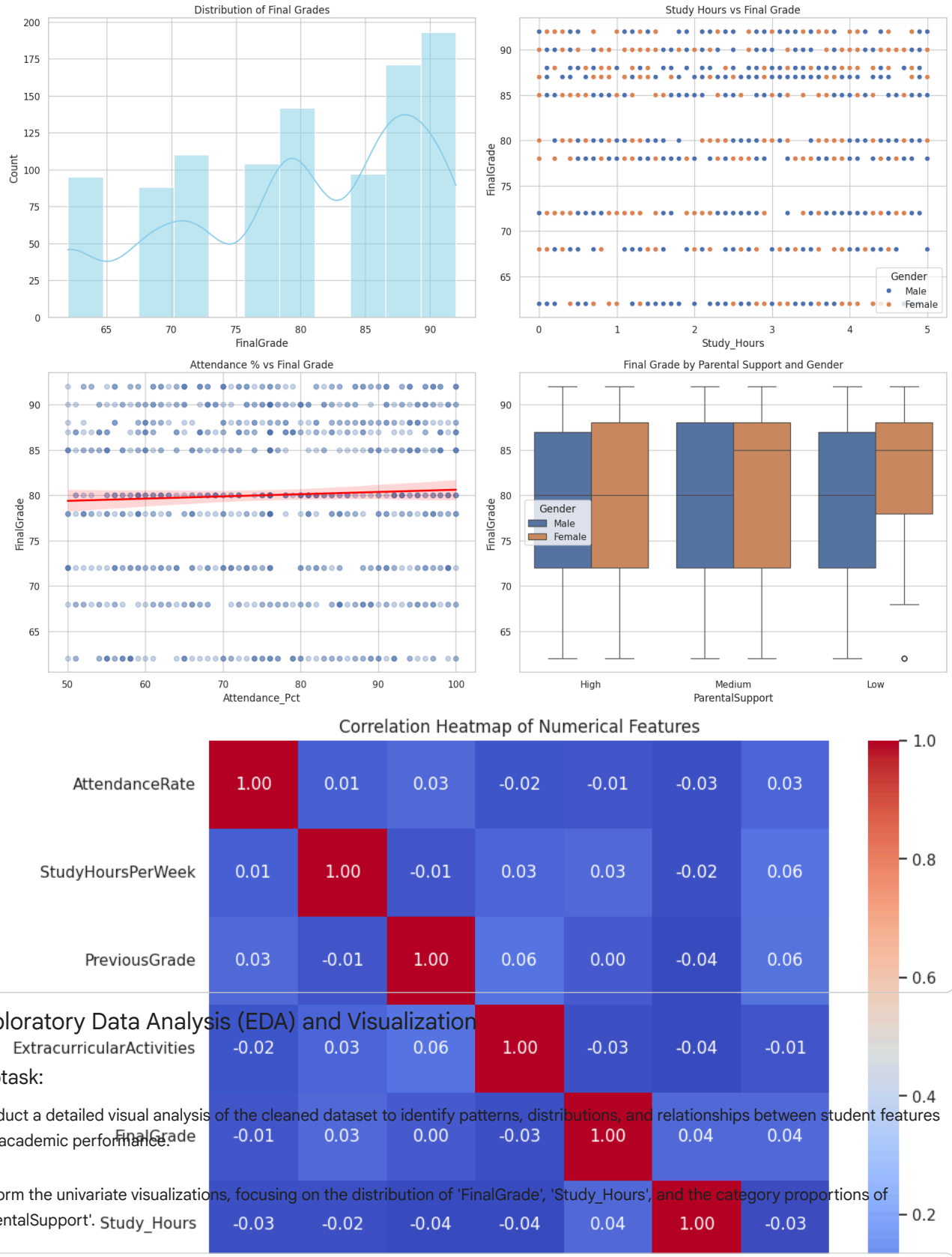
```
# Filter/Cap 'Study Hours' (assuming column 'Study Hours' or 'StudyHoursPerWeek')
# The dataset has both. Applying logic to both to be safe or identifying relevant ones.
if 'Study Hours' in df.columns:
    df.loc[df['Study Hours'] < 0, 'Study Hours'] = 0
if 'StudyHoursPerWeek' in df.columns:
    df.loc[df['StudyHoursPerWeek'] < 0, 'StudyHoursPerWeek'] = 0

# 'Attendance (%)' values that exceed 100
if 'Attendance (%)' in df.columns:
    df.loc[df['Attendance (%)'] > 100, 'Attendance (%)'] = 100
if 'AttendanceRate' in df.columns:
    df.loc[df['AttendanceRate'] > 100, 'AttendanceRate'] = 100

# 4. Drop unnecessary columns
columns_to_drop = ['StudentID', 'Name']
df.drop(columns=[col for col in columns_to_drop if col in df.columns], inplace=True)

# 5. Standardize column names (removing spaces and special characters for ease of use)
df.columns = df.columns.str.replace(' ', '_').str.replace('(', '').str.replace(')', '').str.replace('%', 'Pct')

print('Data cleaning complete. Missing values remaining:', df.isnull().sum().sum())
print('New column names:', df.columns.tolist())
display(df.head())
```

```
Data cleaning complete. Missing values remaining: 0
New column names: ['Gender', 'AttendanceRate', 'StudyHoursPerWeek', 'PreviousGrade', 'ExtracurricularActivities', 'ParentalS
/tmp/ipykernel_283/3227472582.py:15: FutureWarning: Downcasting object dtype arrays on .fillna, .ffill, .bfill is deprecated
  df[col] = df[col].fillna(df[col].mode()[0])
```

	Gender	AttendanceRate	StudyHoursPerWeek	PreviousGrade	ExtracurricularActivities	ParentalSupport	FinalGrade	Study_H
0	Male	85.0	15.0	78.0	1.0	High	80.0	
1	Female	90.0	20.0	85.0	2.0	Medium	87.0	
2	Male	78.0	10.0	65.0	0.0	Low	68.0	
3	Male	92.0	25.0	90.0	3.0	High	92.0	
4	Female	88.0	18.0	82.0	2.0	Medium	85.0	

Going for Exploratory Data Analysis (EDA) subtask to visualize and understand the data distributions and relationships.

```
import matplotlib.pyplot as plt
import seaborn as sns

# Set plotting style
sns.set(style="whitegrid")

# Create a figure with subplots
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# 1. Univariate Analysis: Distribution of FinalGrade
sns.histplot(df['FinalGrade'], kde=True, ax=axes[0, 0], color='skyblue')
axes[0, 0].set_title('Distribution of Final Grades')

# 2. Bivariate Analysis: Study Hours vs Final Grade
sns.scatterplot(data=df, x='Study_Hours', y='FinalGrade', hue='Gender', ax=axes[0, 1])
axes[0, 1].set_title('Study Hours vs Final Grade')

# 3. Bivariate Analysis: Attendance vs Final Grade
sns.regplot(data=df, x='Attendance_Pct', y='FinalGrade', scatter_kws={'alpha':0.3}, line_kws={'color':'red'}, ax=axes[1, 0
axes[1, 0].set_title('Attendance % vs Final Grade')

# 4. Multivariate Analysis: Parental Support, Gender and Final Grade
sns.boxplot(data=df, x='ParentalSupport', y='FinalGrade', hue='Gender', ax=axes[1, 1])
axes[1, 1].set_title('Final Grade by Parental Support and Gender')

plt.tight_layout()
plt.show()

# Correlation Heatmap for numerical features
plt.figure(figsize=(10, 8))
corr = df.select_dtypes(include=['float64', 'int64']).corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap of Numerical Features')
plt.show()
```



Exploratory Data Analysis (EDA) and Visualization

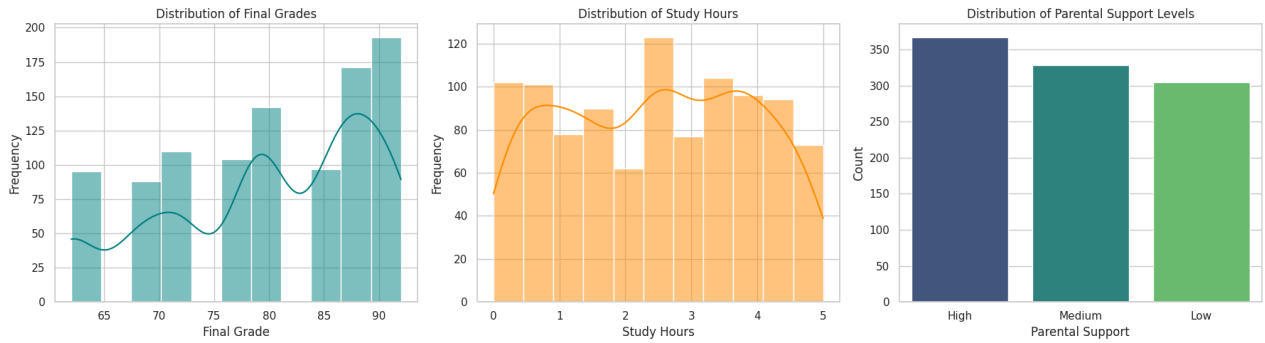
Subtask:

Conduct a detailed visual analysis of the cleaned dataset to identify patterns, distributions, and relationships between student features and academic performance.

Perform the univariate visualizations, focusing on the distribution of 'FinalGrade', 'Study_Hours', and the category proportions of 'ParentalSupport'.

```
import matplotlib.pyplot as plt
import seaborn as sns

# Set the aesthetic style of the plots
sns.set_theme(style="whitegrid")

# Create a figure for univariate analysis
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# 1. Histogram for FinalGrade
sns.histplot(df['FinalGrade'], kde=True, color='teal', ax=axes[0])
axes[0].set_title('Distribution of Final Grades')
axes[0].set_xlabel('Final Grade')
axes[0].set_ylabel('Frequency')
```

```
# 2. Histogram for Study_Hours
sns.histplot(df['Study_Hours'], kde=True, color='darkorange', ax=axes[1])
axes[1].set_title('Distribution of Study Hours')
axes[1].set_xlabel('Study Hours')
axes[1].set_ylabel('Frequency')

# 3. Count plot for ParentalSupport
sns.countplot(data=df, x='ParentalSupport', hue='ParentalSupport', palette='viridis', ax=axes[2], order=df['ParentalSupport
axes[2].set_title('Distribution of Parental Support Levels')
axes[2].set_xlabel('Parental Support')
axes[2].set_ylabel('Count')

plt.tight_layout()
plt.show()
```



Task

Perform bivariate and multivariate exploratory data analysis on the student performance dataset "student_performance.csv", then build and evaluate two predictive models. First, categorize 'FinalGrade' into 'Low', 'Medium', and 'High' and train a Multinomial Logistic Regression model to classify students, evaluating it with Accuracy and a Confusion Matrix. Second, train a Linear Regression model to predict the continuous 'FinalGrade' and evaluate it using R-squared, MAE, and MSE. Conclude by identifying the most significant factors impacting student performance.

Bivariate and Multivariate EDA

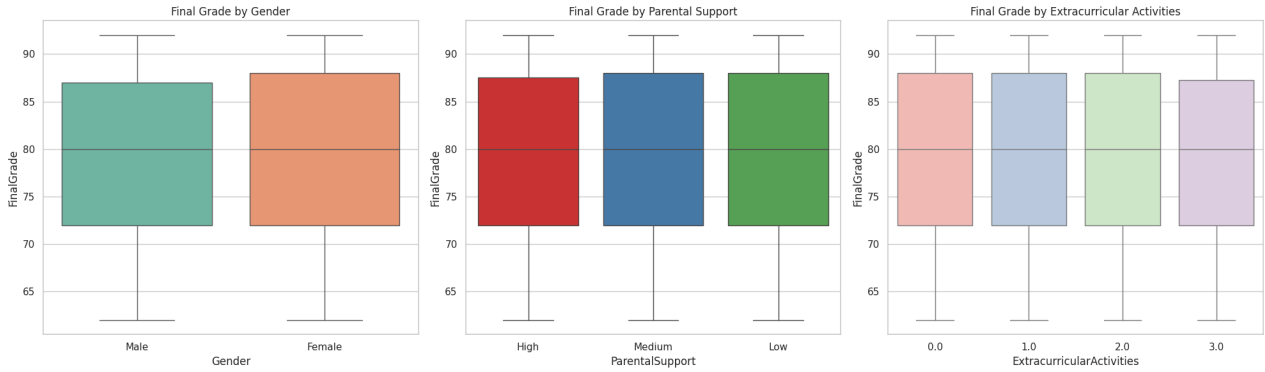
Subtask:

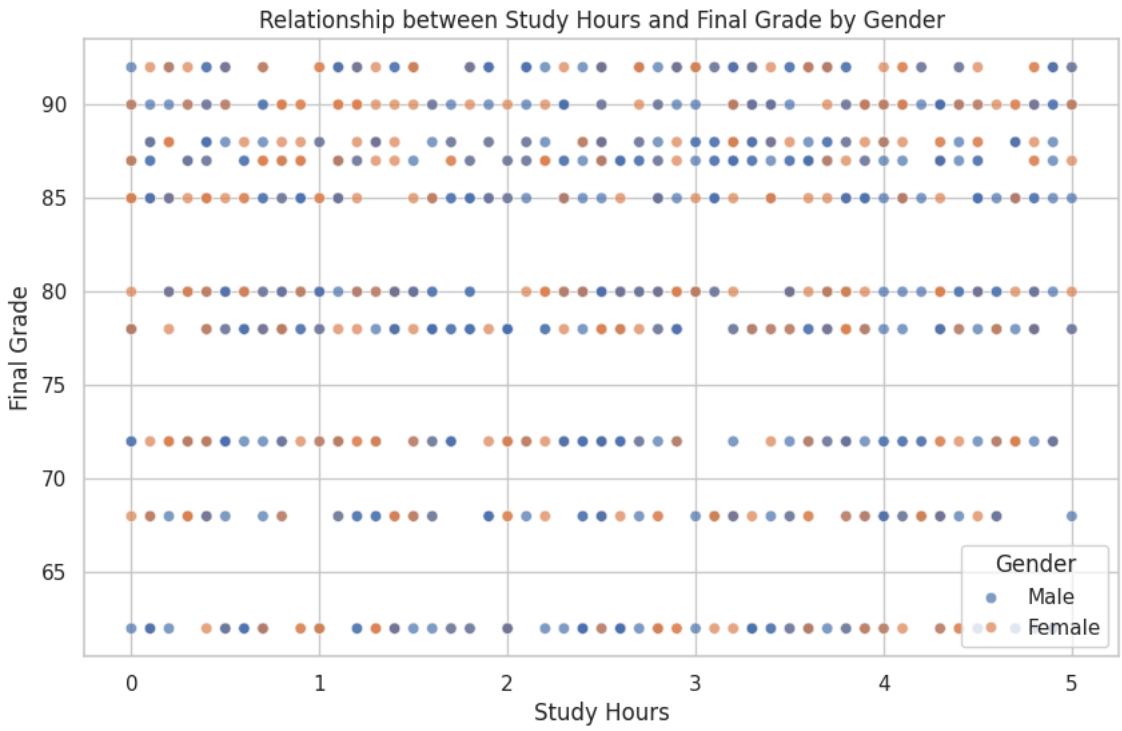
Perform bivariate and multivariate visualizations to explore relationships between student features and 'FinalGrade'.

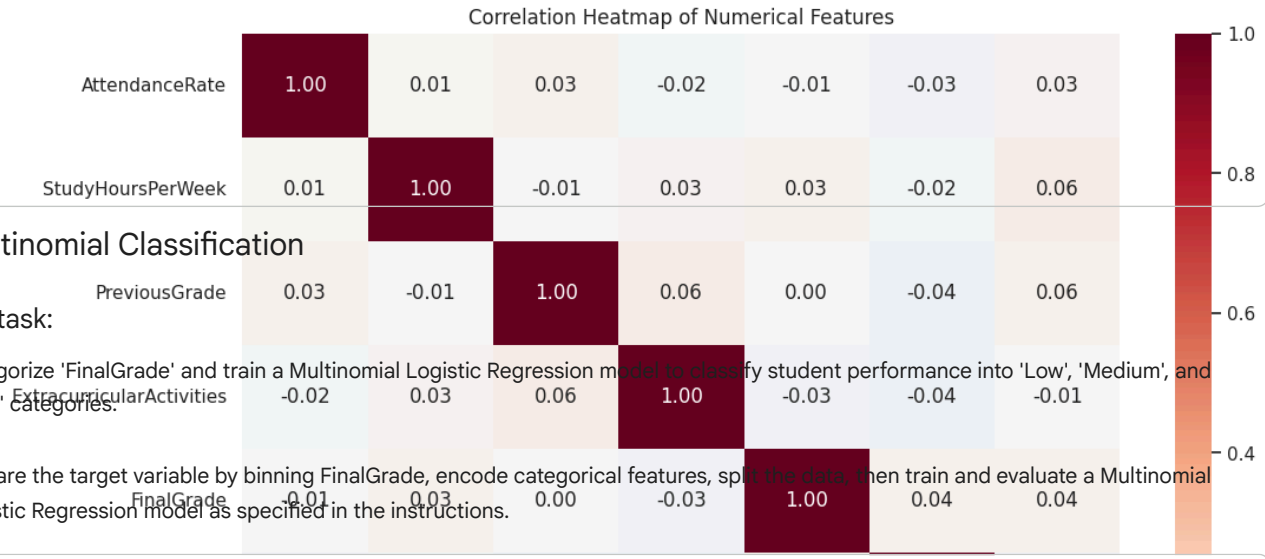
Perform the bivariate and multivariate analysis, including boxplots for categorical variables, a scatter plot for study hours, a correlation heatmap, and a pairplot with parental support as the hue.

```
import matplotlib.pyplot as plt
import seaborn as sns

# Set aesthetic style
sns.set_theme(style="whitegrid")

# 1. Bivariate Analysis: Categorical Features vs Final Grade
fig, axes = plt.subplots(1, 3, figsize=(20, 6))

sns.boxplot(data=df, x='Gender', y='FinalGrade', hue='Gender', palette='Set2', ax=axes[0], legend=False)
axes[0].set_title('Final Grade by Gender')

sns.boxplot(data=df, x='ParentalSupport', y='FinalGrade', hue='ParentalSupport', palette='Set1', ax=axes[1], legend=False)
axes[1].set_title('Final Grade by Parental Support')

sns.boxplot(data=df, x='ExtracurricularActivities', y='FinalGrade', hue='ExtracurricularActivities', palette='Pastel1', ax=
axes[2].set_title('Final Grade by Extracurricular Activities')

plt.tight_layout()
plt.show()

# 2. Scatter Plot: Study Hours vs Final Grade (Multivariate with Gender)
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Study_Hours', y='FinalGrade', hue='Gender', alpha=0.7, palette='deep')
plt.title('Relationship between Study Hours and Final Grade by Gender')
plt.xlabel('Study Hours')
plt.ylabel('Final Grade')
plt.legend(title='Gender')
```

```
plt.show()

# 3. Correlation Heatmap
plt.figure(figsize=(12, 8))
numeric_df = df.select_dtypes(include=['float64', 'int64'])
correlation_matrix = numeric_df.corr()
sns.heatmap(correlation_matrix, annot=True, cmap='RdBu_r', center=0, fmt='.2f')
plt.title('Correlation Heatmap of Numerical Features')
plt.show()

# 4. Pairplot: Subset of Key Features with ParentalSupport as Hue
key_features = ['Study_Hours', 'Attendance_Pct', 'PreviousGrade', 'FinalGrade', 'ParentalSupport']
sns.pairplot(df[key_features], hue='ParentalSupport', palette='viridis', diag_kind='kde', plot_kws={'alpha': 0.6})
plt.suptitle('Pairplot of Key Features by Parental Support', y=1.02)
plt.show()
```







Multinomial Classification

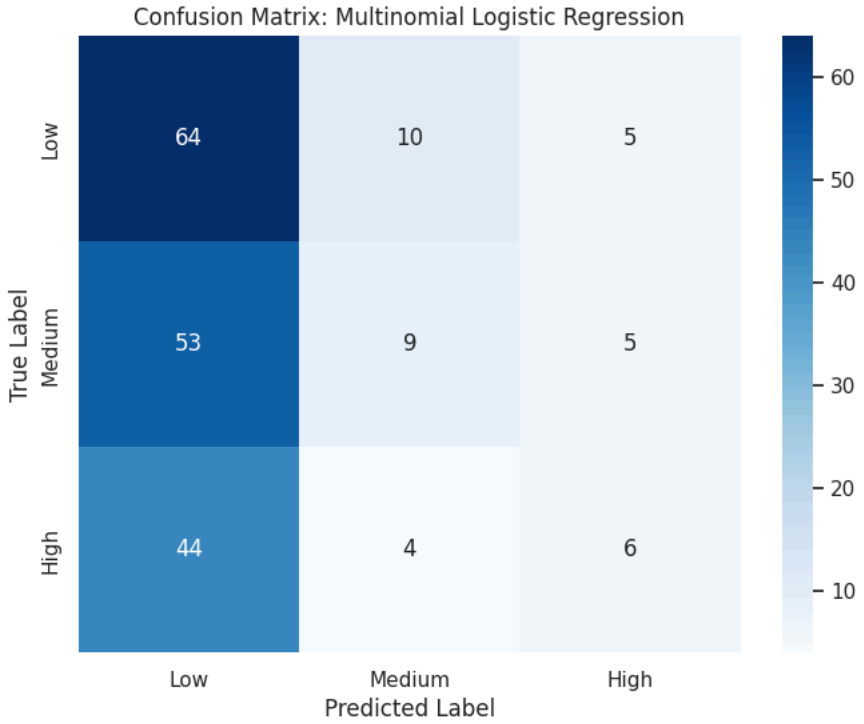
Subtask:

Categorize 'FinalGrade' and train a Multinomial Logistic Regression model to classify student performance into 'Low', 'Medium', and 'High' categories.

Prepare the target variable by binning FinalGrade, encode categorical features, split the data, then train and evaluate a Multinomial Logistic Regression model as specified in the instructions.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# --- 1. Categorize 'FinalGrade' into 'Low', 'Medium', and 'High' ---
# Using quantiles to ensure balanced classes
quantile_labels = ['Low', 'Medium', 'High']
df['Performance_Category'] = pd.qcut(df['FinalGrade'], q=3, labels=quantile_labels)

# --- 2. Encode categorical features ---
le = LabelEncoder()
cat_features = ['Gender', 'ParentalSupport', 'Online_Classes_Taken', 'ExtracurricularActivities']
for col in cat_features:
    # Ensure data is string type before encoding
    df[col] = le.fit_transform(df[col].astype(str))

# --- 3. Split the dataset ---
# Dropping target columns and original FinalGrade
X = df.drop(['FinalGrade', 'Performance_Category'], axis=1)
y = df['Performance_Category']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# --- 4. Feature Scaling (Crucial for Convergence) ---
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# --- 5. Initialize and train Logistic Regression ---
model = LogisticRegression(solver='lbfgs', max_iter=2000, random_state=42)
model.fit(X_train_scaled, y_train)

# --- 6. Predictions ---
y_pred = model.predict(X_test_scaled)

# --- 7. Evaluation ---
accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred, labels=quantile_labels)

print(f'Accuracy Score: {accuracy:.4f}')
print('\nClassification Report:')
print(classification_report(y_test, y_pred))

# --- 8. Visualize Confusion Matrix ---
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=quantile_labels, yticklabels=quantile_labels)
plt.title('Confusion Matrix: Multinomial Logistic Regression')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.show()
```

```
Accuracy Score: 0.3950

Classification Report:
              precision    recall  f1-score   support

        High       0.38      0.11      0.17        54
         Low       0.40      0.81      0.53        79
      Medium       0.39      0.13      0.20        67

    accuracy                           0.40       200
   macro avg       0.39      0.35      0.30       200
weighted avg       0.39      0.40      0.32       200
```



Task

Train a Linear Regression model to predict the continuous 'FinalGrade' variable using the student performance data in "student_performance.csv". Evaluate the model's performance using R-squared, Mean Absolute Error (MAE), and Mean Squared Error (MSE), and include a visualization such as a residuals plot or an actual vs. predicted plot. Finally, provide a comprehensive summary of the project, identifying the most significant factors impacting student performance based on your analysis and models.

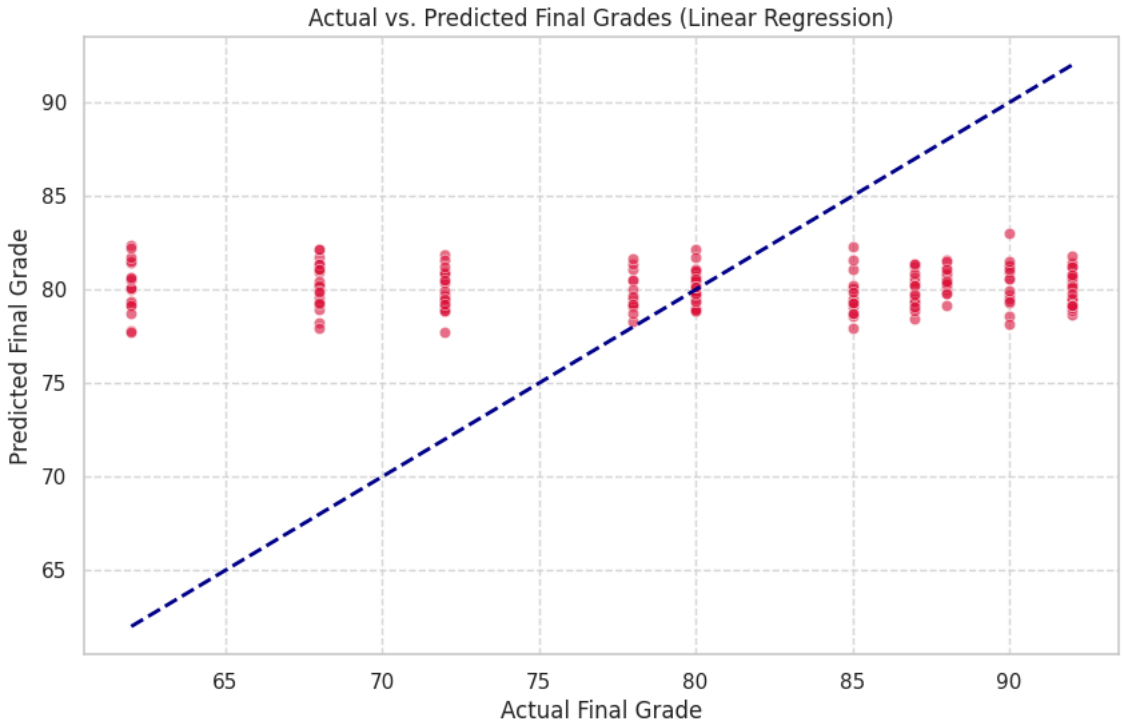
Regression Modeling

Subtask:

Train a Linear Regression model to predict the continuous 'FinalGrade' and evaluate its performance with standard metrics and visualizations.

Train a Linear Regression model using the continuous 'FinalGrade' as the target, evaluate it using R-squared, MAE, and MSE, and visualize the results with a scatter plot of actual vs. predicted values as specified in the instructions.

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Define feature matrix X and target variable y
# Dropping the target columns used for classification and the continuous target itself
X_reg = df.drop(['FinalGrade', 'Performance_Category'], axis=1)
y_reg = df['FinalGrade']

# 2. Split the data into training and testing sets (80/20 split)
X_train_reg, X_test_reg, y_train_reg, y_test_reg = train_test_split(X_reg, y_reg, test_size=0.2, random_state=42)

# 3. Instantiate and train the Linear Regression model
lin_reg = LinearRegression()
```

```
lin_reg.fit(X_train_reg, y_train_reg)

# 4. Make predictions on the test set
y_pred_reg = lin_reg.predict(X_test_reg)

# 5. Calculate evaluation metrics
r2 = r2_score(y_test_reg, y_pred_reg)
mae = mean_absolute_error(y_test_reg, y_pred_reg)
mse = mean_squared_error(y_test_reg, y_pred_reg)

print(f'Linear Regression Performance Metrics:')
print(f'R-squared: {r2:.4f}')
print(f'Mean Absolute Error (MAE): {mae:.4f}')
print(f'Mean Squared Error (MSE): {mse:.4f}')

# 6. Visualization: Actual vs. Predicted Final Grades
plt.figure(figsize=(10, 6))
sns.scatterplot(x=y_test_reg, y=y_pred_reg, alpha=0.6, color='crimson')

# Add a reference line for perfect prediction
max_val = max(y_test_reg.max(), y_pred_reg.max())
min_val = min(y_test_reg.min(), y_pred_reg.min())
plt.plot([min_val, max_val], [min_val, max_val], color='darkblue', lw=2, linestyle='--')

plt.title('Actual vs. Predicted Final Grades (Linear Regression)')
plt.xlabel('Actual Final Grade')
plt.ylabel('Predicted Final Grade')
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```

```
Linear Regression Performance Metrics:
R-squared: -0.0117
Mean Absolute Error (MAE): 8.0279
Mean Squared Error (MSE): 90.8419
```